

Bus Servo Driver Board for Sseed Studio XIAO Datasheet

Product Introduction

The Bus Servo Driver Board supports ST/SC series servos with UART and USB dual modes. It features a 2P 3.5mm screw terminal and is compatible with the Sseed Studio XIAO ecosystem. Ideal for robotic arms and multi-legged robots, it ensures rapid prototyping and precise control.

Product Features

- **Wide Compatibility:** Supports ST/SC series serial bus servos, making it versatile for various robot projects.
- **Flexible Dual Mode Control:** Offers UART and USB dual-mode communication, allowing easy control of multiple servos.
- **Professional Power Supply:** Equipped with a 2P 3.5mm screw terminal for secure and reliable power connections.
- **Compact Design:** Measures only 42mm x 25mm, ideal for integration into space-constrained projects.
- **Ecosystem Support:** Seamlessly integrates with Sseed Studio XIAO series development boards for expanded functionality.

What is the Bus Servo Driver Board?

The Bus Servo Driver Board is a serial bus servo driver designed to manage and control servos operating on serial communication protocols (such as the ST/SC series). Through integrated control circuitry and communication interfaces, it converts user commands into servo actions, enabling coordinated multi-servo operation.

Unlike traditional PWM control, serial bus technology provides more efficient communication and greater scalability, making it particularly suited for robotics projects requiring multiple servos. This driver board offers both UART and USB control options, allowing users to choose between embedded system integration or convenient desktop debugging based on their needs.

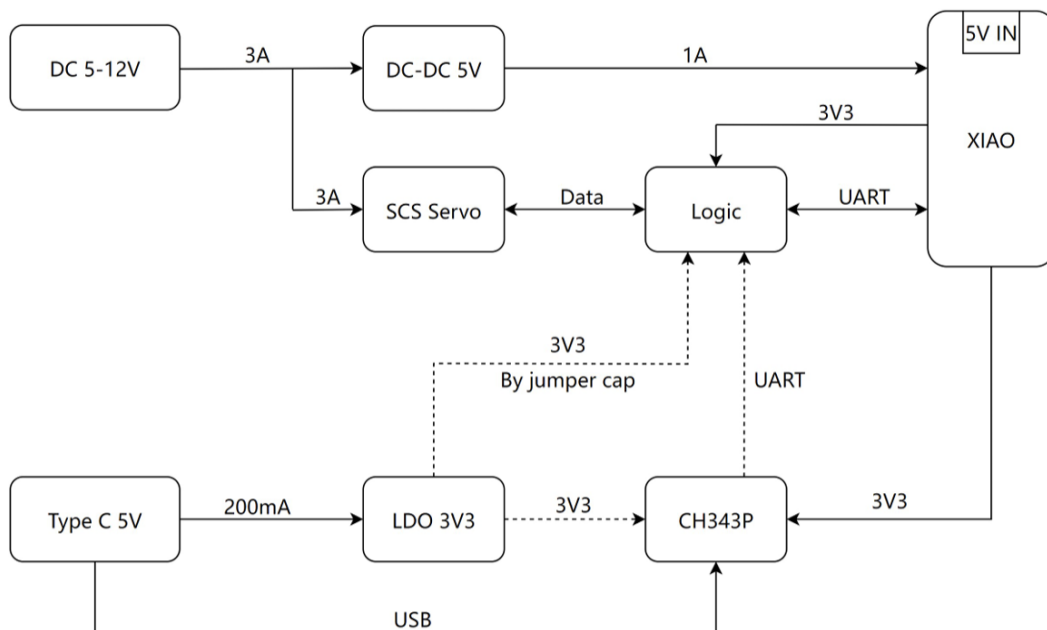
Application Scenarios

- Supports serial bus servo driving for robotic arms.
- Enables servo control in hexapod robots.
- Provides servo driving for humanoid robots.
- Assists wheeled robots with servo support.
- Facilitates servo use in DIY and educational projects.

Specification

Name	Details
Compatible Servo Types	ST/SC series serial bus servos
Servo Control Quantity	Multiple
Control Interface	UART / USB
Communication Protocol	Serial bus protocol (compatible with ST/SC servos)
Input Voltage Range	5-12V DC (dependent on servo voltage)
Power Supply Interface	2P 3.5mm screw terminal
Servo Interface	2.5mm 3P terminal
Jumper Settings	UART/USB mode switching
Dimensions	42mm x 25mm
Weight	15g
Operating Temperature	● 10°C ~ 60°C

Hardware Overview



Documentation

Wiki: https://wiki.seeedstudio.com/bus_servo_driver_board/

PCB:

https://files.seeedstudio.com/wiki/bus_servo_driver_board/202004237_Servo_Driver_Board_for_Seeed_Studio_XIAO_SCH_PDF_250225.pdf

FAQ

Q: How do I power the Bus Servo Driver Board?

A: Connect the power supply via the onboard 2P 3.5mm screw terminal. The input voltage range is 5-12V DC, which must match the rated voltage of the servos used and provide sufficient current to support all connected servos during operation.

Q: How do I connect servos and control devices to the board?

A: Servos are connected to the driver board via the 2.5mm 3P terminal, ensuring proper wiring (power, ground, signal lines). Control devices can be connected to a computer via the USB interface or to a microcontroller via the UART interface, with the specific mode determined by the jumper settings.

Q: How many servos can this driver board control?

A: It can control multiple ST/SC series serial bus servos, with the exact number depending on the power supply capacity and bus load conditions. Theoretically, it supports a large number of servos working together.

Q: What's the difference between USB and UART modes, and how do I choose?

A: USB mode is plug-and-play, ideal for quick debugging and PC-based control; UART mode offers higher real-time performance, suitable for embedded system integration. Choose based on your application: USB for debugging, UART for embedded projects.

Q: What happens if the jumper settings are incorrect?

A: Incorrect jumper settings will result in communication failure. For UART mode, the jumper pads must be soldered; for USB mode, they must remain unsoldered. Adjust using a fine-tipped soldering iron based on the intended mode.